

HE2003 Econometrics I: Linear Regression with Multiple Regressors 3

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Tests and Confidence Intervals in Multiple Regression

KEY CONCEPT: Test of Joint Hypotheses

Consider the following (joint) null hypothesis:

$$H_0 : \beta_1 = 0 \text{ and } \beta_2 = 0.$$

versus the alternative hypothesis:

$$H_1 : H_0 \text{ is wrong (at least one coefficient is nonzero)}.$$

This alternative hypothesis is equivalent to:

$$H_1 : \beta_1 \neq 0 \text{ or } \beta_2 \neq 0 \text{ or both.}$$

The above joint null hypothesis H_0 imposes **two restrictions**. In general, **a joint null hypothesis imposes two or more restrictions on the regression coefficients**. For example, suppose in a regression with $k = 6$ regressors, the null hypothesis is that the **2nd, 4th, and 5th regressors are zero**, then we have $H_0 : \beta_2 = 0, \beta_4 = 0, \beta_5 = 0$, so that there are **3 restrictions**.

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Empirical Application 1: Parents' Education in a Birth Weight Equation

$$bwght_i = \beta_0 + \beta_1 motheduc_i + \beta_2 fatheduc_i + \beta_3 cigs_i + \beta_4 parity_i + \beta_5 faminc_i + u_i,$$

bwght = birth weight, in pounds;

motheduc = years of education for the mother;

fatheduc = years of education for the father;

cigs = average number of cigarettes the mother smoked per day during pregnancy;

parity = the birth order of this child;

faminc = annual family income.

We want to test the hypothesis that **parents' education has no effect on birth weight** after controlling for *cigs*, *parity*, *faminc*. This is stated as $H_0 : \beta_1 = 0$ and $\beta_2 = 0$, and $H_1 : \beta_1 \neq 0$ or $\beta_2 \neq 0$ or **both** (mother's education or father's education or both have effect on birth weight).

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Empirical Application 2: Testing the “Efficient Market Hypothesis” in Financial Economics

- Regression analysis can be used to **test whether the stock market efficiently uses information in valuing stocks.**
- For example, let *return* be the total return from holding a firm's stock over the four-year period from the end of 2019 to the end of 2023.
- The “**efficient market hypothesis**” says that **these return should not be systematically related to information already known in 2019.** If firm characteristics known at the beginning of the period help to predict the stock returns, then we could use this information to make profit. **There should not be such an arbitrage opportunity in a completely efficient market.**

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Empirical Application 2 (continued):

For 2019, let dkr be a firm's debt to capital ratio, let eps denote the earning per share, let $netinc$ denote the firm's net income, and let $salary$ denote the compensation for CEO.

The following equation was estimated:

$$\widehat{return} = -14.37 + 0.321dkr + 0.043eps - 0.0051netinc + 0.0035salary,$$

(6.89) (0.201) (0.078) (0.0047) (0.0022)

$n = 142, R^2 = 0.0395.$

Question: How to test whether none of the regressors with information in 2019 are able to predict the stock return from 2019 to 2023? Overall, is the evidence for predictability of stock returns strong or weak?

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Why Can't We Just Test the Coefficients One at A Time?

- How about using the *t*-statistics and test the restrictions one at a time? (For example, reject $H_0 : \beta_1 = 0$ and $\beta_2 = 0$ if either of the individual t-test (say, t_1 and t_2) exceeds 1.96 in absolute value).
- **Unfortunately, such procedure is incorrect for joint testing:**
The rejection probability under H_0 (Type I error) will not be 5%.

Let t_1 and t_2 be the two t-statistics for β_1 and β_2 , respectively:

$$t_1 = \frac{\hat{\beta}_1 - 0}{SE(\hat{\beta}_1)} \text{ and } t_2 = \frac{\hat{\beta}_2 - 0}{SE(\hat{\beta}_2)},$$

The “one at a time” test will reject H_0 if $|t_1| > 1.96$ or $|t_2| > 1.96$ (reject H_0 if at least one t-test is rejected at 5% level).

Then, what is the Type I error of this procedure? (It should be 5%, if not, this test cannot be used.)

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For simplicity, suppose that the two t-statistics are independent from each other; Then, we have the following result (**need not recite**):

$$\begin{aligned} & P_{H_0} [|t_1| > 1.96 \text{ or } |t_2| > 1.96] \\ &= 1 - P_{H_0} [|t_1| \leq 1.96 \text{ and } |t_2| \leq 1.96] \\ &= 1 - P_{H_0} [|t_1| \leq 1.96] \times P_{H_0} [|t_2| \leq 1.96] \text{ (by independence)} \\ &= 1 - (0.95)^2 \\ &= 0.0975, \end{aligned}$$

which is **not the desired 0.05**. Using similar argument, we have for the joint null hypothesis $H_0 : \beta_1 = 0, \beta_2 = 0, \beta_3 = 0$,

$$1 - P_{H_0} [|t_1| \leq 1.96] \times P_{H_0} [|t_2| \leq 1.96] \times P_{H_0} [|t_3| \leq 1.96] = 1 - (0.95)^3,$$

which is equal to **0.1426**, **much larger** than the desired 0.05.

Solution: Use a different test statistic (the **F-statistic**).

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KEY CONCEPT: The Homoskedasticity-Only F -statistic

There is relatively **simple formula for the F -statistic** that holds under **homoskedasticity**. Despite of its limitation (**it does not work if the error terms are heteroskedastic**), the homoskedasticity-only F -statistic is **useful to understand what the F -statistic is actually doing**.

The homoskedasticity-only F -statistic is given by:

$$F = \frac{(R_u^2 - R_r^2)/q}{(1 - R_u^2)/(n - k - 1)}.$$

where

- R_u^2 : the R^2 from the **unrestricted regression**,
- R_r^2 : the R^2 from the **restricted regression** (under restrictions in H_0),
- q is the **number of restrictions** in H_0 , and k the **number of regressors in the unrestricted regression** (intercept not included in k).

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In particular, the R^2 from the unrestricted regression is:

$$R_u^2 = \frac{ESS_u}{TSS} = \frac{\sum_{i=1}^n (\hat{Y}_{u,i} - \bar{Y})^2}{\sum_{i=1}^n (Y_i - \bar{Y})^2},$$

where $\hat{Y}_{u,i}$ is the OLS predicted value under the unrestricted regression.

And the R^2 from the restricted regression is:

$$R_r^2 = \frac{ESS_r}{TSS} = \frac{\sum_{i=1}^n (\hat{Y}_{r,i} - \bar{Y})^2}{\sum_{i=1}^n (Y_i - \bar{Y})^2},$$

where $\hat{Y}_{r,i}$ is the OLS predicted value under the restricted regression, which uses the q restrictions imposed in H_0 .

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The “Restricted” and “Unrestricted” Regressions

Example 1: Parents' Education and Baby's Birth Weight

In the example of baby's birth weight, the null hypothesis that **“parents' education has no effect on birth weights”** (after controlling for the other regressors) and the alternative hypothesis that **parents' education does matter** are

$$H_0 : \beta_1 = 0 \text{ and } \beta_2 = 0$$

(or we can just write $H_0 : \beta_1 = \beta_2 = 0$)

versus

$$H_1 : \text{either } \beta_1 \neq 0 \text{ or } \beta_2 \neq 0 \text{ or both}$$

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The **Restricted Regression Model** can be written as:

$$bwght_i = \beta_0 + \beta_3cigs_i + \beta_4parity_i + \beta_5faminc_i + u_i. \quad (1)$$

The **Unrestricted Regression Model** can be written as:

$$bwght_i = \beta_0 + \beta_1motheduc_i + \beta_2fatheduc_i + \beta_3cigs_i + \beta_4parity_i + \beta_5faminc_i + u_i. \quad (2)$$

- The number of restrictions under H_0 is $q = 2$.
- The number of regressors under the unrestricted regression is $k = 5$.
- Therefore, the F -statistic for this problem can be written as:

$$F = \frac{(R_u^2 - R_r^2)/2}{(1 - R_u^2)/(n - 6)},$$

where R_r^2 is the R^2 computed from eq.(1), and R_u^2 from eq.(2).

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Example 2: Do school resources matter?

Let $Expn$ denote **expenditures per student** and consider the multiple regression model:

$$TestScore = \beta_0 + \beta_1 STR + \beta_2 Expn + \beta_3 PctEL + u.$$

Then, the null hypothesis that **“school resources do not matter”** and the alternative that they do matter correspond to

$$H_0 : \beta_1 = 0 \text{ and } \beta_2 = 0$$

(or we can just write $H_0 : \beta_1 = \beta_2 = 0$)
versus

$$H_1 : \text{either } \beta_1 \neq 0 \text{ or } \beta_2 \neq 0 \text{ or both}$$

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The **Restricted Regression Model** can be written as:

$$TestScore = \beta_0 + \beta_3 PctEL + u. \quad (3)$$

The **Unrestricted Regression Model** can be written as:

$$TestScore = \beta_0 + \beta_1 STR + \beta_2 Expn + \beta_3 PctEL + u. \quad (4)$$

- The number of restrictions under H_0 is $q = 2$.
- The number of regressors under the unrestricted regression is $k = 3$.
- Therefore, the F -statistic for this problem can be written as:

$$F = \frac{(R_u^2 - R_r^2)/2}{(1 - R_u^2)/(n - 4)},$$

where R_r^2 is the R^2 computed from eq.(3), and R_u^2 from eq.(4).

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- By construction, $F \geq 0$ (One can show that R^2 increases with the number of regressors, so that R_u^2 is always larger than R_r^2).
- If R_u^2 is much larger than R_r^2 , then the two regressors *STR* and *Expn* probably do have explanation power for *TestScore* ($\beta_1 \neq 0$ and/or $\beta_2 \neq 0$). This can be understood as the evidence against $H_0 : \beta_1 = \beta_2 = 0$.
- In other words, the F-statistic is comparing the fittings of the two regressions; if the “unrestricted” model fits the data sufficiently better than the “restricted” model, we will reject the “restricted” model (the model under the joint null hypothesis H_0).
- Just as we compare the t-statistic with the Normal distribution to determine the critical values for the t-test, we need to compare the F-statistic with certain distribution to find the critical values for the F-test. Here, we need to use the Chi-squared distribution.